MILITARY SPECIFICATION

CARTRIDGE CASE, 20MM, EMPTY, PRIMED

This Specification is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions and preparation for delivery for the cartridge case designated as Cartridge Case, 20mm, Empty, Primed, for use in propellant testing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard, Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-147	-	Palletized Unit Loads Validated Oct 1974

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

U.S. Army Armament Research & Development Center (ARDC)

PRODUCT AND PACKAGING DRAWINGS

9381752 - Cartridge Case, 20mm, Empty, Primed
9386921 - Packing and Marking of Fiberboard Box, for Interplant Shipment: Cartridge Case, 20mm, Empty, Primed

INSPECTION EQUIPMENT

7296962 - Indicator Gage
11075284 - Primer Resistance Test Circuit

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publications. The following documents form a part of this specification to the extend specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR-100-199 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Article

(The Interstate Commerce Commision Regulations are now part of the Code of Federal Regulations (1967 editions and revisions) available from the superintendent of documents, Governmenr Printing Office, Washington, DC 20315. Orders for the above publications, cite 49CFR-100-199 (Latest revision).)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge case. The cartridge case, empty, primed, shall comply with all requirements specified on drawing (dwg) 9381752, all associated drawings and with all requirements specified in applicable specifications.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Electrical resistance of primed case (M52A3B1 primer only). The electrical resistance of the primed case shall not be less than 500 ohms or more than 1,200,000 ohms.

3.5 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.5.1 Processing defects. The metal parts and assemblies shall be free of cracks, splits, ruptures, perforations, dents, deformities, exposed explosive and other forms of substandard workmanship.

3.5.2 Cleanliness. All components and assemblies shall be free of foreign matter, such as oil, grease, corrosion, dirt, chips, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (4.3)

b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

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4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Cartridge Case, 20mm, Empty, Primed	9381752	200

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. Rejection of the first article sample shall be in accordance with Table I and MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70552 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9381752
	Cartridge Case, 20mm, Empty, Primed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Case, 20mm, Empty, Primed (Dwg 9381752)				
	Examination for defects	200	Acc-0 Rej-1	3.2	4.4.2.1
	Primed Cartridge Case (M52A3B1 primer only)				
	Electrial Resistance	200	Acc-0 Rej-1	3.4	4.4.2b/4.5.1

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. A lot shall consist of cartridge cases manufactured by one manufacturer in one unchanged process in accordance with the same drawings and drawing revisions and the same specification and specification revisions. Each inspection lot shall consist of primers and cartridge cases from one interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Primed cartridge cases (M52A3B1 Primer only). Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.5.1. Primed cases which fail to meet the requirements of 3.4 shall be removed.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70552 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.1	Cartridge Case, 20mm, Empty, Primed 1/		1 OF 2	9381752
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Electric resistance of primed case (after primer crimping and waterproofing - electric primer only)		100%	3.4	4.4.2b/4.5.1
102.	Cocked, loose, missing or inverted primer		.25%	3.2	Visual/Manual
103.	Depth of primer seating		.25%	3.2	7296962
104.	Corrosion		.25%	3.2	Visual
105.	Head configuration		.25%	3.2	Visual
106.	Metal sliver(s) on case head (electric primer)		.25%	3.2	Visual
107.	Case vent seal missing (when required)		.25%	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9381752
4.4.2.1	Cartridge Case, 20mm, Empty, Primed 1/				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR:					
201.	Lacquer on primer button (electric primer)		.40%	3.2	Visual
202.	Primer crimp missing or incomplete		.40%	3.2	Visual
203.	Waterproofing lacquer missing around primer		.40%	3.2	Visual
204.	Dents or scratches		.40%	3.5	Visual
205.	Presence of foreign matter except corrosion		.40%	3.5	Visual
206.	Evidence of poor workmanship		.65%	3.5	Visual

NOTES:
1/ See MIL-STD-651 for defect standards.

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4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in paragraph 4.4.2.1 under the Inspection Method column. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Methods of inspection.

4.5.1 Electrical resistance of primed case (M52A3B1 primer only). The primed case shall be tested for the applicable requirement using Government approved equipment and circuitry specified on Dwg. 11075284 to measure the resistance of the primer.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level B. (See 5.2.1).

5.2 Packing.

5.2.1 Level B. The cartridge cases shall be packed in accordance with Dwg. 9386921.

5.2.2 Unitized loads.

a. Palletization shall be in accordance with MIL-STD-147.

b. For CONUS or interplant shipment, commercial expendable pallets, 40" x 48" (4 way) may be used in lieu of those specified in 5.2.2.a. Weight capacity of the pallet selected shall be adequate for the load. Gross weight of load shall not exceed 1500 pounds and height, including pallet, shall not exceed 41 inches.

5.3 Marking. Marking shall be in accordance with Dwg. 9386921.

5.4 Shipping. When cartridge cases from more than one lot are shipped at one time each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The cartridge case covered by this specification is intended for use in propellant testing.

6.2 Ordering data. See MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, ATTN: AMSMC-QAF-I (D), Dover, NJ 07801-5001. Request letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, or US Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AB25)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70552	CARTRIDGE CASE, 20MM, EMPTY, PRIMED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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